

# Governing Open Source as National Infrastructure

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### Executive Summary

Open source software is now foundational infrastructure covering national digital services, cybersecurity capabilities, cloud platforms, developer tooling, and increasingly AI systems. All such core technologies depend on open source software developed and maintained in global ecosystems. Yet most governments still manage this dependency informally, with open source responsibilities fragmented across agencies, security oversight is uneven, and procurement treats open source software tactically rather than strategically. Furthermore, contributions to upstream critical open source projects such as core Digital Public Infrastructure (DPI) systems are rare and uncoordinated. This mismatch between dependency and governance fosters real risk because open source software use is unmanaged at the level its importance demands.

This article makes a neutral, analytical case for why governments should establish national Open Source Program Offices, or national OSPOs. It explains what a national OSPO is, what challenges it addresses, and how it functions as institutional infrastructure rather than an advocacy initiative. The article also outlines the key focus areas where national OSPOs are becoming essential, including cybersecurity, software supply chains, open source AI, public digital infrastructure, and digital sovereignty.

The central argument is that open source software is already embedded in governments' critical systems, and national OSPOs are

how governments govern that reality deliberately, consistently, and at scale.

## Introduction

Governments around the world are deep into their digital transformation journey, with many of the public services now software-driven, policy implementation increasingly depends on source code, and national resilience includes digital resilience and calls for digital sovereignty. At the same time, the software foundations of government have quietly shifted. Open source software is embedded in almost every critical system that governments operate. This reliance on open source software happened gradually. Frameworks, platforms, libraries, operating systems, databases, cloud platforms, and AI tooling are either open source software or rely on open source software, often hundreds per system (think of digital ID, digital payment, and data exchange systems).

While the dependency of such systems on open source software has become systemic, governance has not, and it is lagging. Most governments still treat open source software as a technical detail handled by individual teams, or as a procurement checkbox addressed late in the process (exactly how the case was with enterprises 25 years ago). That approach no longer matches reality as it creates security blind spots, economic inefficiencies, and long-term strategic exposure.

This article explains why open source software should be understood and treated as national digital infrastructure, why ad hoc management fails at the national scale, and why national OSPOs are emerging as a necessary governance response. The goal is not to promote open source ideology, but to describe a practical institutional model that governments can use to reduce risk and increase strategic control over systems they already depend on.

Supporting material:

- Why Governments Need OSPOs in Support of Digital Public Infrastructure
- Leveraging Open Source to Build Digital Public Infrastructure

## Open Source a National Infrastructure

Open source is already embedded across government systems. It runs identity services, tax platforms, healthcare systems, transportation infrastructure, data analytics pipelines, developer environments, and cloud-native platforms. Increasingly, it also underpins AI frameworks, model tooling, and data processing pipelines used by the public sector. The United States' national strategy to secure global leadership in AI (Winning the Race: America's AI Action Plan - July 2025) provides open source AI with unprecedented visibility in federal strategy. Read my analysis on the topic in this [LinkedIn article](#).

Independent studies consistently show that the vast majority of modern software systems depend on open source software, and governments are not an exception. States are now dependent on global, distributed software ecosystems that they do not directly control to deliver essential public services. That dependency is not inherently negative, as some might promote. In fact, it brings transparency, adaptability, and innovation speed (key aspects of the open source development model) that proprietary models often cannot match. However, dependency without governance is exposure and risk.

- When no one has a complete view of what software is used where, security incidents cascade slowly.
- When government agencies act independently, public money is spent repeatedly on solving the same problems.
- When expertise is outsourced by default, governments lose leverage and optionality over time.

These outcomes are not caused by open source software. They are caused by the absence of institutional ownership. We’ve learned these lessons in the enterprise work, and it is time for us to apply those lessons in the public sector.

Supporting material:

- For a deep dive on “Strategic Open Source Lessons from Industry Leaders”, please visit this [LinkedIn post](#) that includes pointers to over a dozen articles on the topic published in 2025.

## The Governance Gap

In most governments, open source responsibility is scattered, and just like in enterprises, legal teams worry about licenses, security teams worry about vulnerabilities, procurement teams worry about contracts, and developers worry about delivery. No one is accountable for the whole system. As a result, decisions are local while consequences are national. Private-sector organizations faced this problem years ago, actually decades ago, since the early 2000s. As open source became mission-critical, leading companies established OSPOs to centralize policy, manage risk, coordinate contributions, and engage upstream communities. Governments now face the same structural challenge, but at much larger scale and with higher stakes.

## Defining a National OSPO

At its core,

A national OSPO is a permanent institutional capability with a clear mandate to govern how the state uses, contributes to, and invests in open source software.

This is not new. If I am to define what an enterprise OSPO is, the definition would be slightly different: an **enterprise** OSPO is a permanent institutional capability with a clear mandate to govern how the **company** uses, contributes to, and invests in open source software. uses, contributes to, and invests in open source software.

A national OSPO is not a developer forum, a compliance-only function, or a temporary digital transformation project. Its authority is horizontal, cutting across ministries, agencies, and sectors. It does not replace existing teams working with open source software, instead it coordinates them. A well-designed national OSPO provides a single point of accountability for open source strategy at the national level.

## Core Functions of National OSPOs

While organizational models vary, effective national OSPOs converge on a common set of functions. These functions reflect the minimum capabilities required to manage open source software at a national scale.

- **Policy and governance:** A national OSPO establishes clear, practical policies for open source use and contribution, including guidance on licensing, contribution rules for public servants, and default positions for publicly funded software. The goal is consistency, as good policies reduce friction and remove uncertainty.
- **Security and supply chain risk management:** Open source security is often a coordination problem. A national OSPO enables visibility into where open source components are used, supports vulnerability tracking across agencies, and coordinates response with national cybersecurity authorities. This setup dramatically improves response time and reduces systemic risk.
- **Procurement and economic coordination:** A national OSPO enables shared assessments and reference architectures. It shifts procurement conversations to long-term sustainability, exit options, and resilience.
- **Upstream engagement and contribution:** Governments are major consumers of open source software, whether they acknowledge it or not. A national OSPO coordinates contributions, supports maintenance of critical projects (for example, those used in building DPI systems), and ensures that government requirements are addressed upstream rather than through costly downstream workarounds.
- **Capacity Building:** National OSPOs invest in training developers, lawyers, procurement officers, and managers. They build institutional memory that survives individual projects and political cycles.

## Key Focus Areas for National OSPOs

As government systems evolve, several domains make national OSPOs increasingly critical.

- **Cybersecurity and software supply chains:** Supply chain attacks have made dependency visibility a national security concern. National OSPOs provide the structure to manage exposure systematically and respond quickly when vulnerabilities emerge across multiple agencies.
- **Open Source AI governance:** AI systems increasingly depend on open models, frameworks, and datasets. Without open source governance, governments risk losing transparency, accountability, and long-term control over AI systems they deploy. National OSPOs connect AI policy objectives with the realities of open source AI ecosystems.
- **Public Digital Infrastructure:** Digital identity, registries, data platforms, and payment systems are long-lived, critical national assets. When these systems are built on open source, governance becomes as important as architecture. OSPOs help ensure that infrastructure remains adaptable and vendor-independent over decades.
- **Digital sovereignty:** Open source enables governments to understand and govern the systems they depend on. National OSPOs institutionalize that capability.
- **Operating at a national scale:** There is no single correct OSPO structure - a true statement for an enterprise OSPO as well. Some governments will centralize authority, and others will adopt federated or hybrid models. What remains important is the mandate, authority, and integration with existing processes. The failure mode is failing to assign ownership of open source governance.

## Conclusion

Open source software already runs the state. Governments did not choose this deliberately, but they now live with the consequences. The only remaining decision is whether they govern this reality strategically or continue managing it reactively. National OSPOs are a structural response to a structural dependency. For governments serious about cybersecurity, resilience, digital sovereignty, and long-term accountability, making the case for a National OSPO is no longer theoretical or optional. The path forward for governments does not require radical change.

- First, acknowledge open source as critical infrastructure.
- Second, establish a national OSPO with a clear mandate and stable funding.
- Third, start small, demonstrate value, and scale deliberately.

The worst option is to continue treating open source governance as an afterthought. The opportunity exists to replace reactive management with intentional stewardship.

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